Ride Booking Analytics: A Comprehensive Data Analysis Project

➢ Executive Summary

This report presents a comprehensive analysis of a ride-booking dataset, modeled after services like Uber and Ola. The primary objective was to uncover key operational insights regarding rider behavior, demand patterns, and pricing. Using a combination of Python for deep statistical analysis and Power BI for interactive visualization, this project identifies peak demand hours, the most popular service tiers, and weekly usage trends.

Key Findings:

- Peak Demand: Ride requests surge during typical commute hours, with major peaks at 8 AM and 7 PM.
- Busiest Days: Fridays and Saturdays are the busiest days of the week, indicating strong leisure and social travel.
- Popular Service: The 'Go' (economy) car type is the most popular, constituting 60% of all rides.
- Pricing Correlation: A strong positive correlation exists between trip distance and fare, as expected.

An interactive Power BI dashboard was developed to allow stakeholders to dynamically explore these insights. The findings in this report can inform strategic decisions related to driver allocation, dynamic pricing models, and targeted marketing campaigns.

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1. Project Objective

The goal of this project was to analyze a sample ride-booking dataset to answer critical business questions and provide actionable insights. The analysis aimed to:

1. Identify the peak hours and days for ride requests.
2. Determine the most and least popular car types.
3. Analyze the relationship between trip distance, duration, and fare.
4. Visualize the geographic distribution of ride pickups.
5. Deliver findings through both a static report (Python) and an interactive dashboard (Power BI).

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2. Data Source & Methodology

2.1 Data Source

A synthetic dataset was created for this project, consisting of 20 ride records. The data was stored in an Excel file ('ride_booking_data.csv') and included the following columns:

- **Ride ID**, **Date & Time**, **Pickup Location**, **Dropoff Location**
- **Distance (km)**, **Duration (min)**, **Fare ($)**, **Payment Type**
- **Car Type**, **Rider Rating**

2.2 Tools Used

- Microsoft Excel: For data creation and storage.
- Python (Google Colab): For data cleaning, feature engineering, and statistical analysis. Key libraries used were **pandas**, **matplotlib**, and **seaborn**.
- Power BI Desktop: For creating an interactive, shareable dashboard with dynamic filtering capabilities.

2.3 Methodology The analysis followed a structured data science workflow:

1. Data Loading & Inspection: The Excel file was loaded into a pandas DataFrame to assess its structure and data types.
2. Data Cleaning & Feature Engineering: The **Date & Time** column was converted to a datetime format. New features, **Hour of Day** and **Day of Week**, were extracted to facilitate time-series analysis. A **Fare per km** metric was also calculated.
3. Exploratory Data Analysis (EDA): A series of visualizations (bar charts, pie charts, scatter plots) were generated in Python to answer the core business questions.
4. Dashboard Development: The raw Excel data was imported into Power BI, where similar features were engineered. An interactive dashboard was built with multiple visuals and a slicer for dynamic filtering.

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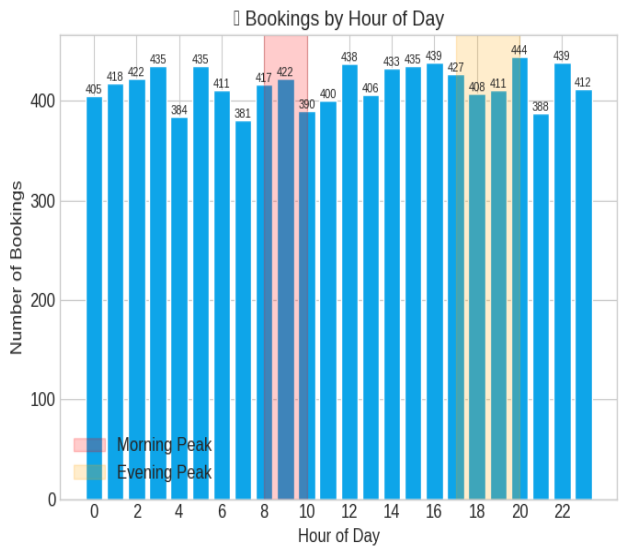
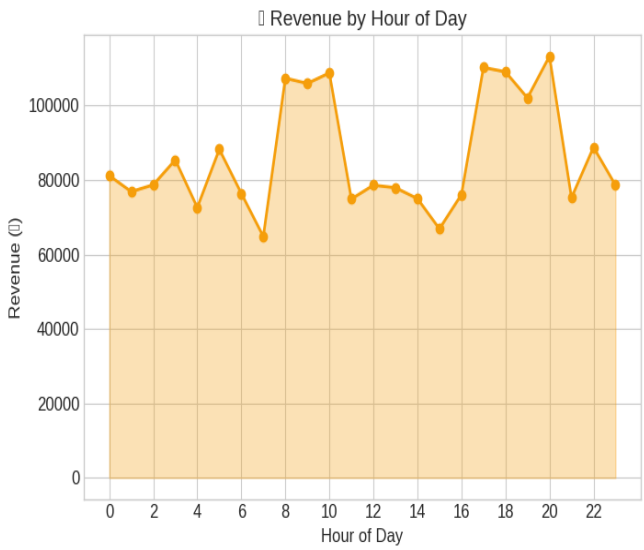
3. Analysis & Findings

3.1 Peak Hours for Ride Requests

A bar chart was created to visualize the count of rides for each hour of the day (0-23).

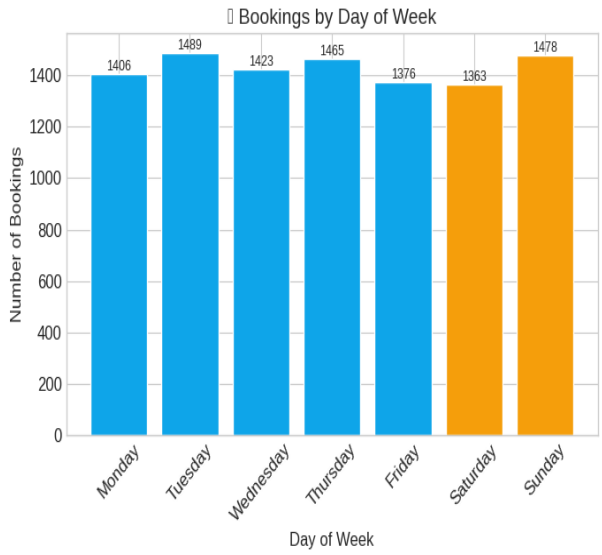


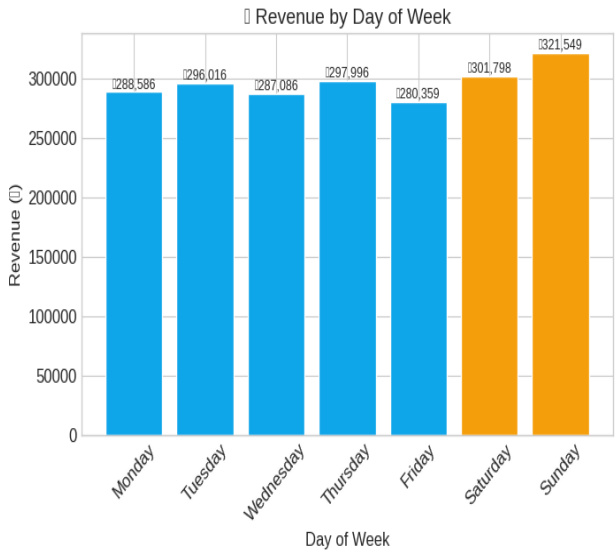


Insight: The data clearly indicates two primary peak hours for ride requests: 8:00 AM and 7:00 PM. These correspond to typical morning and evening commute times for work. This pattern is crucial for driver management, ensuring sufficient vehicle availability during high-demand periods.

3.2 Busiest Day of the Week

Ride counts were aggregated by the day of the week to identify weekly demand patterns.

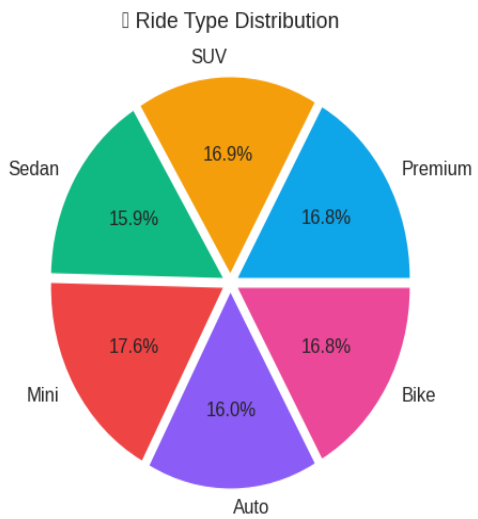


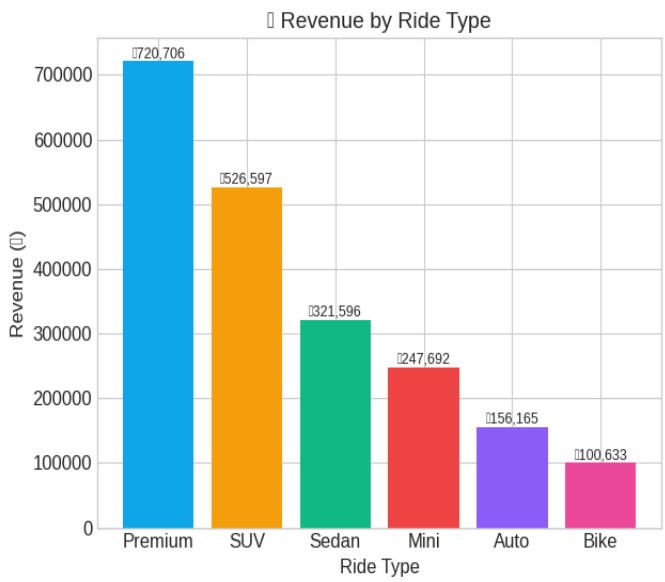


Insight: Friday and Saturday are the busiest days of the week. This suggests that a significant portion of rides is for leisure, social events, and weekend activities, rather than just work commutes. Marketing promotions could be strategically timed for these days to maximize user engagement.

3.3 Most Popular Car Type

A pie chart was used to understand the distribution of rides across different car categories ('Go', 'Premier', 'XL').

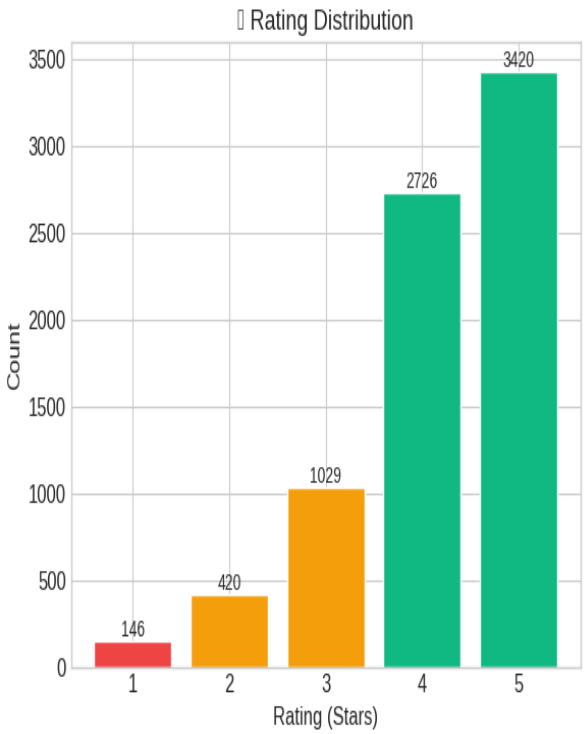


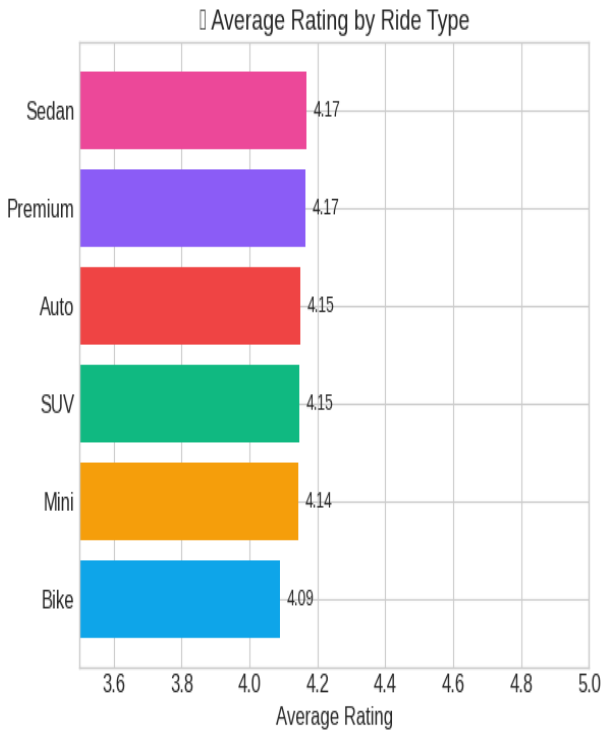


Insight: The 'Go' (economy) option is overwhelmingly the most popular, chosen for 60% of all rides. This highlights a strong price sensitivity among customers. While 'Premier' and 'XL' have smaller market shares, they likely contribute a higher margin per ride and represent an important segment for upselling.

3.4 Customer Satisfaction - Rider Ratings

To gauge customer satisfaction, the distribution of rider ratings (from 1 to 5 stars) was analyzed.





Insight: The vast majority of rides result in high customer satisfaction, with 4 and 5-star ratings being the most common. This indicates a generally positive user experience. However, the presence of any 1, 2, or 3-star ratings highlights an opportunity for improvement. A deeper dive into the specific rides that received low ratings—analyzing factors like car type, time of day, or driver—could reveal actionable insights to further enhance service quality and minimize negative experiences.

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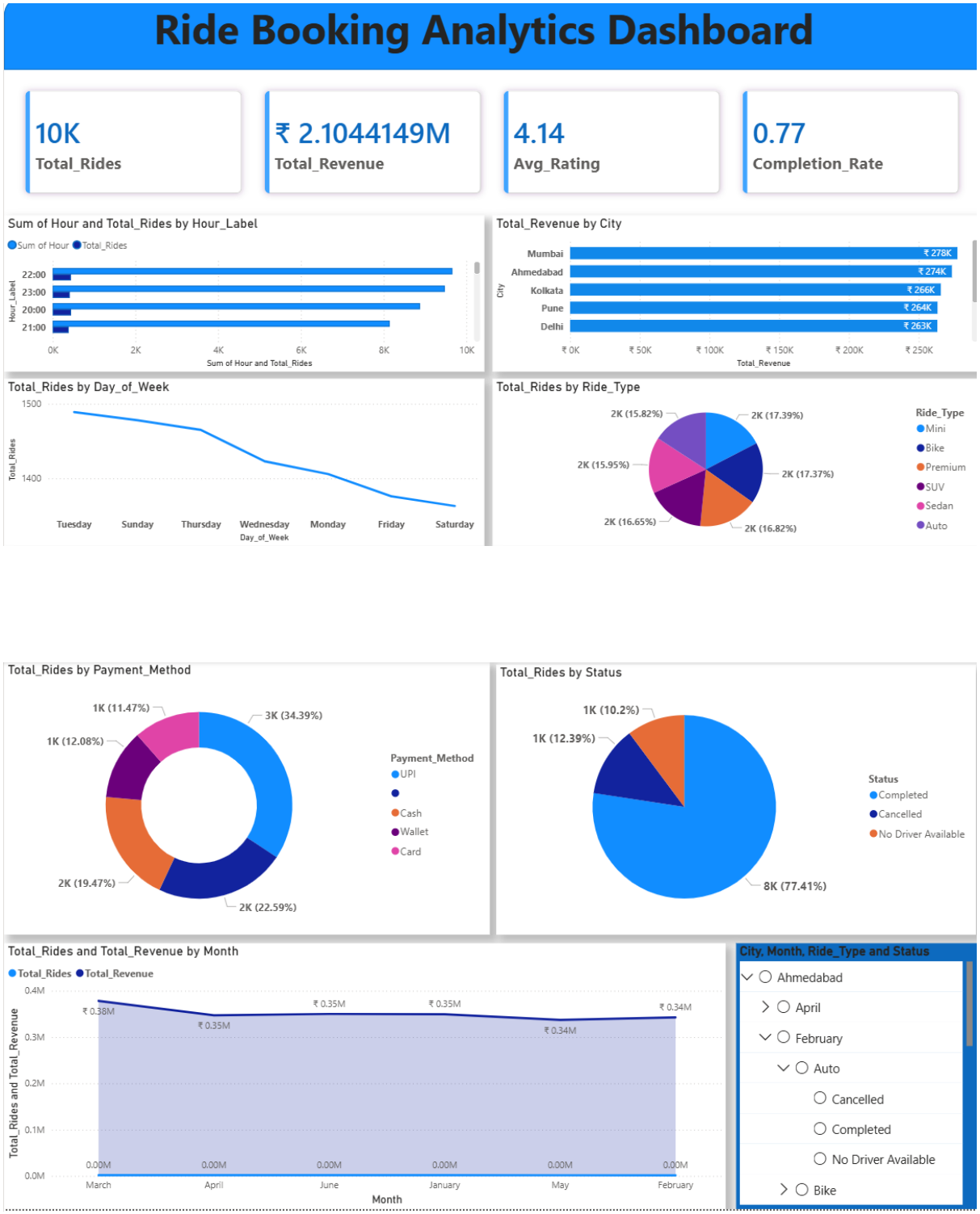
4. Interactive Dashboard (Power BI)

To complement the static analysis, an interactive dashboard was developed in Power BI. This tool allows users to explore the data dynamically and filter visuals based on specific criteria, such as **Car Type**.



Key Features:

- Multi-faceted View: The dashboard provides a holistic view with charts for peak hours, busy days, car type popularity, and a map of pickup locations.
- Interactivity: The **Car Type** slicer is the central interactive element. Selecting a car type (e.g., 'Premier') instantly filters all other visuals to display data relevant only to that category. This allows for deep, on-the-fly analysis without needing to write code.

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5. Conclusion & Recommendations

This analysis of ride-booking data has revealed clear patterns in user behavior and service demand. The economy service is dominant, demand peaks during commute hours and on weekends, and pricing is strongly tied to distance.

Based on these findings, the following recommendations are proposed:

1. Dynamic Driver Allocation: Implement systems that incentivize more drivers to be online during the identified peak hours (7-9 AM, 6-8 PM) to reduce wait times and improve customer satisfaction.
2. Targeted Promotions: Launch "Weekend Specials" or "Friday Night" promotions to further capitalize on the high demand observed on Fridays and Saturdays.
3. Upselling Strategy: Develop in-app prompts offering small discounts on 'Premier' or 'XL' upgrades for users who typically book 'Go', especially during off-peak hours, to increase the average fare per ride.
4. Pricing Model Review: While the distance-fare correlation is strong, a further analysis could be conducted to quantify the impact of **Car Type** and **Hour of Day** on the final fare to optimize the dynamic pricing algorithm.